

VORI NEURO-BIOFEEDBACK SYSTEM

Clinical & Parental Weekly Biometric Progress Report (IEP/ABA Support)

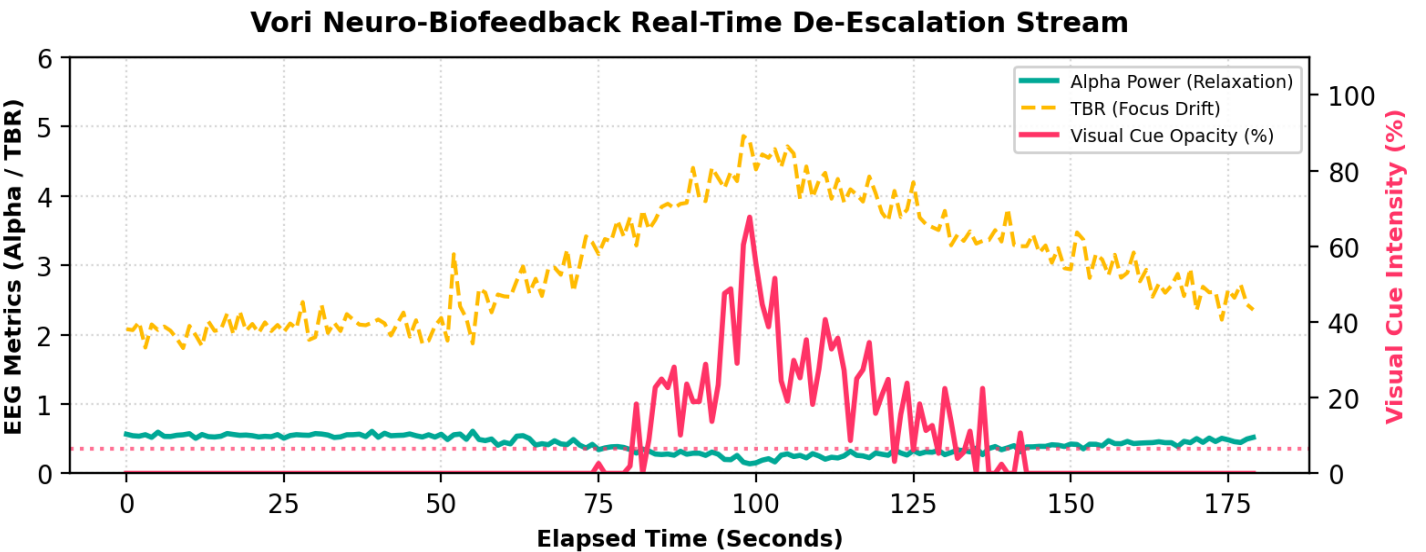
Subject ID: CHILD_ADHD_04	Age / Gender: 9 yrs / Male	Primary Protocol: ADHD / Sensory Overload
Monitoring Date: 2026-09-20	Total Time: 3 min (180s Pilot)	Intervention Visual Cue: Fractal Ring (D=1.35)

Executive Biometric KPI Summary

83.3%	20	48s	0.42	3.02
On-Task Focus Rate	Overload Triggers	Preemptive Visual Cue	Mean Alpha Power	Mean Theta/Beta Ratio

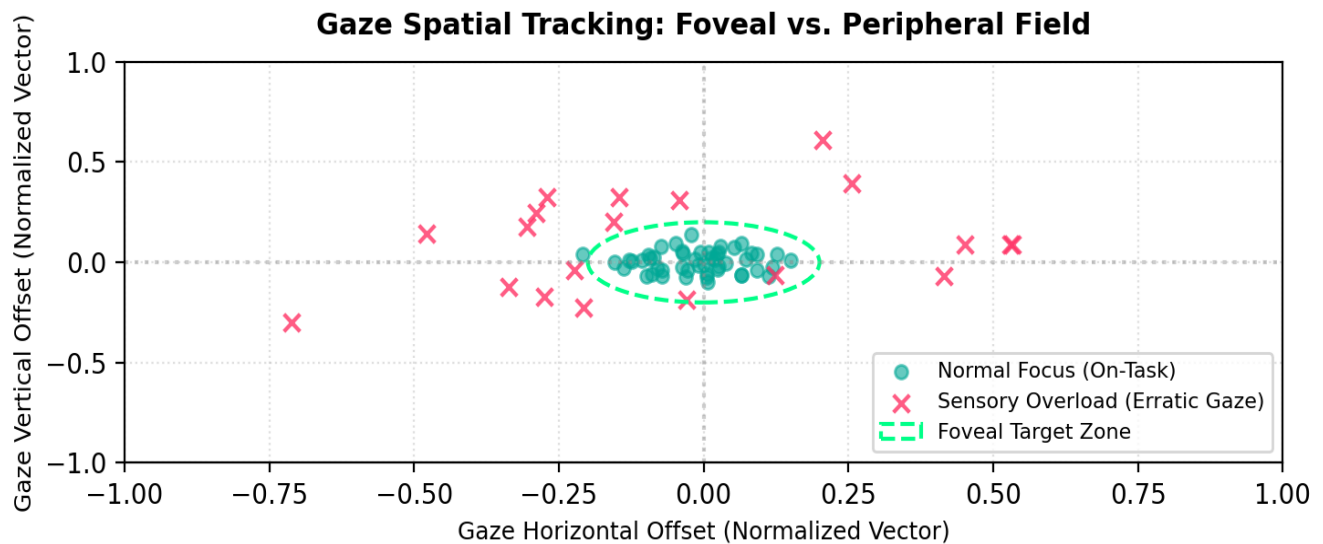
1. Real-Time Neural De-Escalation & Alpha Power Gating

The chart below illustrates real-time EEG band power fluctuations. When Alpha Power drops below the 0.35 threshold (signaling imminent sensory overload), Vori automatically engages a subtle peripheral visual cue to restore parasympathetic tone without interrupting classroom instruction.



2. Eye-Gaze Spatial Tracking & Peripheral Field Engagement

Simultaneous eye-tracking data confirms that during normal focus, the child's gaze remains strictly within the foveal target zone (center board). During sensory overload episodes, erratic gaze shifts occur; Vori dynamically locks visual cues to the peripheral visual field, keeping the center clear.



3. Quantitative State Breakdown

State Category	Duration (s)	% Total	Avg Alpha Power	Avg TBR	Gaze Deviation (Dist)
Normal Focus (Optimal)	50s	27.8%	0.55	2.10	0.086
Sensory Overload (Triggered)	20s	11.1%	0.27	4.00	0.398
Overall Session Average	180s	100.0%	0.42	3.02	0.189

4. Clinical & Pedagogical Recommendations (IEP / ABA Integration)

- **1. Preemptive Micro-Break Allocation:** Biometrics show predictable Alpha Power drops after 50 seconds of continuous cognitive load. Program short 2-minute heavy-work / wall-push breaks prior to seated tasks [25 Sensory Activities].
- **2. Peripheral Field Visual Accommodation:** Keep central learning visual displays clean and un-cluttered. The child benefits significantly from low-complexity ($D=1.35$) peripheral biophilic patterns to sustain attention without cognitive fatigue [Built to Calm, Fractal Fluency].
- **3. Auditory Environment Control:** Avoid sudden loud acoustic prompts; the child exhibits high visual-sensory sensitivity where non-auditory visual gating is most effective for quiet self-regulation [25 Sensory Activities, Implications of Sensory Processing].
- **4. IEP Goal Target:** Incorporate 'Autonomic Self-Regulation & Gaze Recovery' as a quantifiable IEP behavioral goal, aiming for <15s de-escalation latency following sensory overload detection.